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MILESTONE 2 (MIDTERMS) — Dart Logic Practical

(Mobile Development w/ Lab)

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Reflection Paragraph (5–8 Sentences)

This Dart application was architected around an abstract contract, **ElectronicDevice**, to enforce mandatory initialization behavior across all hardware implementations. The base class, **Cellphone**, establishes a shared foundation for mobile operations while strictly enforcing encapsulation by wrapping private fields like **_speaker** with validated getters and setters to protect internal state integrity. Specialized subclasses— **Smartphone** and **FeaturePhone —** leverage method overriding **(override)** to achieve dynamic polymorphism, allowing distinct devices to execute custom camera and audio routines while maintaining a unified interface within List and Map collections. To avoid bloating the vertical inheritance tree, the **WirelessCharging** mixin was composed horizontally using the with keyword, granting specialized capabilities only to compatible devices. Recognizing that mobile applications frequently communicate with external servers, asynchronous methods returning **Future** types were integrated to handle simulated network delay via **Future.delayed.** Using **async** and **await** ensures background processes—such as downloading firmware updates and syncing cloud backups—execute without blocking the main event loop or freezing application execution. Outputting console logs immediately before awaiting promises demonstrates non-blocking behavior, proving that program flow remains responsive during background tasks. Ultimately, combining core object-oriented principles with asynchronous execution produces a clean, scalable, and maintainable codebase aligned with modern mobile development standards.